SIGNAL | Autonomous Financial Research

Computed, not retrieved.

SIGNAL discovers financial signals by finding **persistent breaches of three expectations** — a company's own history, its peers, and its management's narrative — and reports a breach only after it **survives a pre-registered attempt to falsify it**. *Find the breach. Trace the evidence. Try to break the hypothesis.*

01	02	03	04	05	06	07
EXPECT	**BREACH**	**INVESTIGATE**	**TRIANGULATE**	**CHALLENGE**	**SURVIVE**	**DISCOVER**
self · peer · narrative	persistent, on drivers	mechanism + consequence	independent families	9 pre-registered tests	alternatives fail	auditable to the line

01 THE PROBLEM

A non-obvious finding is, almost by definition, not written down anywhere. Retrieval can only return what someone has already said.

DASHBOARDS	retrieve known metrics — headline ratios, not the drivers beneath them.
SUMMARISERS	retrieve known narratives; a "critic" prompt argues — it does not test.
THE GAP	No one systematically computes where **history ≠ peers ≠ narrative**, then tries to explain it away before calling it a finding.

02 THE MECHANISM — THREE EXPECTATIONS

expectation vs observed = breach

SELF	What the company's own history predicts for each decomposed driver
PEER	What same-archetype peers show in the same quarters — removes sector and macro effects
NARRATIVE	What management said drives the numbers — attributions and guidance, timestamped

EXPECTATION — what history, peers and management imply **VS** OBSERVED — reconciled numbers from primary filings **=** BREACH — a candidate — not yet a discovery

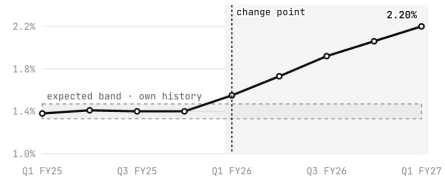
03 FROM CANDIDATE TO DISCOVERY — EVERY GATE MUST PASS

the candidate generator is only the beginning

G1	G2	G3	G4	G5	G6
Reconciled numbers	**Corroborated**	**Company-specific**	**Falsification survived**	**Testable consequence**	**Temporally consistent**
Facts satisfy accounting identities.	≥2 independent source families.	Survives same-quarter peer residuals.	9 alternatives, tested as code.	Predicts data it was not built from.	Persistent; timing coheres.

SELF — expectation 01

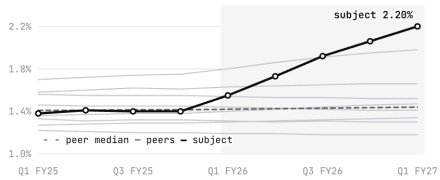
What does its own history predict?



Expected slippage (pre-window band)	1.33–1.47%
Observed, Q1 FY27	2.20%
Breach	+0.80 pp · 5Q outside band

PEER — expectation 02

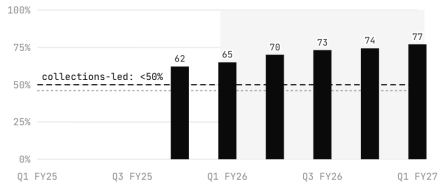
What do comparable banks show, same quarters?



Peer median (IQR)	1.44% (1.32–1.59)
Subject, Q1 FY27	2.20%
Breach	+0.76 pp · 6 of 8 peers flat

NARRATIVE — expectation 03

What did management say drives the numbers?



Claimed driver (5 of 6 calls)	collections
Implied write-off share	< 50%
Breach · observed share	71% (peer median 46%)

EXPECTATION: Slippage ~1.40% · peers ~1.44% · improvement collections-led **VS** OBSERVED: Slippage 2.20% · 71% of NPA reductions are write-offs **=** BREACH · ALL THREE EXPECTATIONS: Improvement is write-off-led while new stress rises — company-specific, persistent, unnarrated.

Prototype · three-expectation view of a discovery

Illustrative prototype data · synthetic panel

04 WHY IT IS DIFFERENT

sophisticated where the problem demands it — simple everywhere else

Computed, not retrieved

Candidates come from accounting-identity driver trees and a management claim ledger — never from mining arbitrary correlations.

The LLM never types a number

Every figure is a rendered reference to a computed node; the language model reads text and writes digit-free prose.

Independent evidence families

Company, peer filings, regulator, rating agency. Syndicated news collapses to one — and is never sole evidence.

Counter-hypotheses tested as code

Sector, regulatory, accounting, denominator, one-off, data and mechanism alternatives — each with a fixed test.

Narrative vs numbers, in time

Bitemporal facts: when the change began, when it became visible, when management's language moved, when the press noticed.

Full provenance + published funnel

Number → formula → inputs → document → page → passage. Every rejected candidate is shown with its reason.

Screens: prototype on simulated data — not findings about any company

SIGNAL | Company in → auditable discovery out · live target 5–7 min

05 RESEARCH WORKFLOW

code by default · the language model only where language is the problem

01 code	02 code+llm	03 code+llm	04 code	05 code	06 code	07 code	08 llm
INGEST	**NORMALIZE**	**EXPECT**	**BREACH**	**TRIANGULATE**	**FALSIFY**	**RANK**	**DISCOVER**
Exchange filings, XBRL, transcripts, RBI, ratings	Typed facts, page anchors, reconciliation	History, 8 peers, claim ledger	Robust surprise, persistence	Counted by source family	9 tests + corollaries	Gates, score, strength rule	8 elements, digit-free prose

06 DISCOVERY FUNNEL

none hand-picked

143	Candidate breaches	any expectation, any driver
31	Structurally valid	reconciled · ≥2Q · no discontinuity
11	Independently corroborated	≥2 families
4	Peer-specific	residualised
2	Survived falsification	9 tests
1	**High-confidence discovery**	strength: Strong

Rejected: sector-wide · one-off · weak evidence · already disclosed · denominator · artefact

ILLUSTRATIVE COUNTS · SIMULATED RUN

07 TRY TO BREAK THE DISCOVERY

pre-registered tests

AE1	**Sector-wide effect**	Company vs same-quarter peer distribution
AE2	**Regulatory / macro**	Difference vs peers around dated sector events
AE3	**Accounting artefact**	Policy diff; merger and restatement flags
AE4	**Denominator effect**	Lagged ratios strip growth dilution
AE5	**One-off / window**	Specification curve across windows
AE6	**Already disclosed**	Claim ledger + press-coverage search
AE7	**Data integrity**	Reconciliation; second-document check
AE8	**Alternative mechanism**	Corollaries on unused data
AE9	**Counter-evidence**	Bounded search, tier allow-list

STRONG	MODERATE	WEAK
≥3 families · ≥4Q · peer-specific · no fail, ≤1 partial	≥2 families · ≥3Q · one alternative partly explains	anything else · always labelled

DISCOVERY 01 OF 1 · ASSET QUALITY · BANK ARCHETYPE

Over six quarters the subject bank's gross NPA ratio fell from 2.31% to 1.62% — but 71% of all NPA reductions were write-offs, while its fresh-slippage ratio rose from 1.40% to 2.20% as the peer median stayed flat at 1.44%. Management credited collections in 5 of 6 earnings calls.

The headline asset-quality improvement is real but is not what management says it is: it is write-off-led, not collections-led, and it coincides with a company-specific rise in fresh slippages that peers do not share.

✓ DISCOVERY · ✓ WHY IT MATTERS · ✓ EVIDENCE · ✓ TIMELINE · ✓ NUMBERS · ✓ COUNTERARGUMENT · ✓ FALSIFICATION CONDITION · ✓ EVIDENCE STRENGTH

EVIDENCE STRENGTH

STRONG

Falsification tests survived	8 / 9
Independent source families	3
Persistence	5 quarters
Peer-specific	outside IQR 4 of 4Q
Visibility gap	4 quarters
Candidates considered	143

Prototype · discovery view — every underlined claim opens its evidence trail

Illustrative output · not a finding about any company

08 EVIDENCE — NUMBER TO PASSAGE

01 DERIVED	**+₹14,953 cr** fresh slippages, 6 quarters
02-03 FORMULA	open + slippages - upgrades - recoveries - write-offs = close 11,550 + 14,953 - 1,892 - 2,906 - 11,748 = 9,957 ✓
04 DOCUMENT	Results — notes to accounts · Tier 1 · hashed
05 PASSAGE	**Additions — fresh slippages (1 of 6 quarters) 3,266** Reductions — write-offs (2,534)

09 TEMPORAL REASONING

illustrative

QUARTER	Q1·25	Q2	Q3	Q4	Q1·26	Q2	Q3	Q4	Q1·27
NUMBERS	1.38	1.41	1.40	1.40	**1.55↑**	**1.73↑**	**1.92↑**	**2.06↑**	**2.20↑**
NARRATIVE				coll.	coll.	coll.	coll.	coll.	**shift**
PEERS	1.41	1.41	1.42	1.42	1.42	1.42	1.43	1.44	1.44

EVENT · CHANGE DETECTED · COVERED

visibility gap · 4 quarters

10 GENERALIZATION

archetype, not company

BANKS	NBFCs	INSURERS	INFRA/HOLDCO
NPA flows, deposits	AUM, funding, Stage 2	APE, VNB, persistency	generic identities

PRACTICE 3 · **HELD-OUT NIFTY** 9 · **NON-NIFTY** 8 · **AS-OF** backtests

11 STACK · DATA · OUTPUT

DATA	NSE/BSE filings (XBRL results, annual reports, LODR transcripts), RBI, rating rationales, peers. Public only.
STACK	Python · DuckDB provenance · XBRL + PDF extraction · FTS5 · LLM at temp 0, cached
OUTPUT	Interactive dashboard, click-through lineage, PDF export. No recommendation.

12 BUILD PLAN

now: prototype on simulated data

08–12 Oct	Ingestion + reconciled extraction (banks, NBFCs)
11–15 Oct	Driver & peer panels, claim ledger, candidates
14–19 Oct	Falsification, provenance, live dashboard
19–23 Oct	Benchmark; insurer + generic modules
23–25 Oct	Hardening, failover, live rehearsals

13 MAPPED TO THE JUDGING CRITERIA

Discovery	Evidence	Novelty	Temporal	Cross-source	Falsifiability	Explainability	Robustness	Generality
three-way breaches	primary, reconciled	narrative + visibility gap	change point, lag	≥2 families	9 tests + condition	lineage + funnel	peer residuals	archetype logic

Prototype figures simulated · benchmark results reported at finals, not claimed here